



Task Plan

Prototype → Production MVP

How to take the validated STAR OnePlatform prototype to a deployable, production MVP across four teacher- and leadership-facing capabilities.

Document	1 of 3 — Task Plan
Date	June 10, 2026
Basis	Build telemetry through commit 7af2303
Team	Tech Lead (you) + 3 developers + 1 QA analyst
Target	Deployable production MVP (4 capability areas)
Estimated timeline	~8 months (range 7–9)

CONFIDENTIAL — Prepared for internal planning & deployment approval. Estimates are rough-order-of-magnitude planning figures, not commitments.

1. Executive summary

The prototype has already retired the hardest technical risk: a working, event-driven teacher loop in which an offline-captured assessment session syncs, flows over a Kafka event backbone, and automatically advances curriculum in a separate service — verified live and in a real browser. This plan takes that prototype to a deployable production MVP covering four capabilities, with a five-person team, in approximately eight months.

The MVP is deliberately scoped. One capability (Assignment & Progress) is essentially proven and needs hardening; two (Onboarding & Setup, Admin & Leadership View) are partially built and need production identity, roster sync, and a reporting surface; one (Planning & Orchestration) is net-new but bounded to rules-and-heuristics for the MVP. Surrounding all four is the production-readiness work — cloud platform, security, FERPA compliance, CI/CD, and monitoring — detailed in Document 3.

MVP capability scope:

MVP capability	Scope	Status	Production gap
1 • Teacher Onboarding & Setup	Sign-in, roster sync, classroom context, initial setup	PARTIAL	Real IAM/SSO, live Clever/ClassLink sync + dedup, MFA, setup flows
2 • Planning & Orchestration	Identify needs, surface next-best actions, suggest schedule	NET-NEW	New orchestration service: needs engine + scheduling over progress data
3 • Assignment & Progress	Assign activities, view student progress, connect data to platform	VALIDATED	Harden & scale the proven teacher loop; broaden activity types
4 • Admin & Leadership View	Usage visibility, operational dashboard, reporting	PARTIAL	New reporting/analytics service + usage telemetry + dashboards

VALIDATED proven in the prototype • PARTIAL partly built • NET-NEW to be built.

Integrated predictive model (delivered). A scikit-learn logistic-regression model predicts each student's likelihood of meeting their IEP goals by annual review and surfaces it as color-coded risk — green / yellow / red — on student records and a district-wide leadership roll-up (it scores live from the data spine via a dedicated Python service). Built and validated in the prototype; the production task below adds a retraining pipeline, drift monitoring, and model governance. It directly powers the Planning/Orchestration and Admin/Leadership capabilities.

2. Major workstreams

ID	Workstream	Scope	Primary owners
WS-A	Production platform foundation	Terraform landing zone, EKS, CI/CD, observability, secrets/security baseline	Tech Lead + Dev 1
WS-B	Onboarding & identity (Cap 1)	IAM/SSO (Cognito, Clever/ClassLink, SAML/LTI), roster sync + dedup, setup flows	Dev 1 + Dev 2
WS-C	Assignment & progress (Cap 3)	Productionize the teacher loop; scale events; activity types; offline sync gateway	Dev 2 + Dev 3
WS-D	Planning & orchestration (Cap 2)	Needs-identification + next-best-action (rules) + schedule suggestions	Dev 2 + Tech Lead
WS-E	Admin & leadership view (Cap 4)	Usage telemetry, operational dashboards, reporting service & exports	Dev 3
WS-F	Security, privacy & FERPA	Encryption, authz, secrets, FERPA controls, pen test, SOC 2 readiness	Tech Lead (cross-cutting)
WS-G	Quality, accessibility & UAT	Test automation, e2e/contract, WCAG, load testing, pilot UAT	QA Analyst + all
WS-H	Predictive model (MLOps)	Productionize scoring service; scheduled retraining, drift & performance monitoring, model registry, bias/fairness review	Tech Lead + Dev 1

3. Sequencing & timeline

Eight-month plan for the five-person team. Foundation and identity lead; the proven Assignment/Progress loop is hardened early to free capacity for the net-new Planning/Orchestration and Admin work; security, FERPA and QA run continuously and gate launch.

Workstream	M1	M2	M3	M4	M5	M6	M7	M8
WS-A · Platform foundation								
WS-B · Onboarding & identity (C1)								
WS-C · Assignment & progress (C3)								
WS-D · Planning/orchestr. (C2)								
WS-E · Admin & leadership (C4)								
WS-F · Security / FERPA								
WS-H · Predictive model (MLOps)								
WS-G · QA, a11y & UAT								
Pilot UAT → production deploy								

■ primary build · ■ continuous / ramp · MVP production deploy at end of M8.

4. Role allocation

Role	Responsibilities
Tech Lead / Principal Eng (you)	Architecture, cloud platform & infra, security & FERPA, IAM design, orchestration logic, code review, release owner
Developer 1 — Backend / Data	Roster sync connectors, data-engine hardening, DB-per-service split, event backbone, service APIs
Developer 2 — Full-stack	Onboarding/setup flows, Assignment/Progress hardening, Planning/Orchestration engine
Developer 3 — Frontend / Full-stack	Web shell, Admin/Leadership dashboards & reporting UI, offline/PWA, accessibility implementation
QA Analyst	Test strategy & automation, e2e/contract tests, accessibility (axe), load-test coordination, UAT

5. Milestones

#	Milestone	Exit criteria
M1	Production foundation live	CI/CD commit→prod, IaC landing zone, observability + secrets baseline
M2	Identity + first live roster	Real SSO; a pilot district roster syncs from Clever/ClassLink into the graph (Cap 1 alpha)
M3	Hardened teacher loop	Assignment/Progress productionized in staging under load (Cap 3)
M4	Planning/Orchestration alpha	Next-best-action + schedule suggestions over progress data (Cap 2)
M5	Admin/Leadership dashboards	Usage + operational reporting feature-complete (Cap 4)
M6	Security & compliance gate	FERPA controls, encryption/secrets, pen test & WCAG audit passed
M7	Pilot UAT	One district end-to-end on production infra; defect burn-down
M8	Production MVP deploy	Go-live to initial cohort; on-call, runbooks, status page in place

6. Key assumptions

- The validated event backbone and teacher loop are reused as-is — no architectural redesign.
- Clever/ClassLink sandbox access and signed data-sharing agreements (DPAs) are available by week 2.
- A committed pilot district with a known roster source is identified before M2.
- “Next-best-action” for the MVP is rules/heuristics over existing progress data — not a trained ML model.
- Brand and accessibility design tokens are delivered early in the engagement.
- MVP launches to a pilot cohort first, then ramps toward the 2,500-district / 25,000-teacher initial target.

7. Key risks

Risk	Sev.	Mitigation
Lean team vs. four capabilities + hardening	High	Strict scope discipline; reuse the proven loop; defer non-MVP activity types; consider fractional security/design help
Planning/Orchestration is net-new	Med	Constrain to rules/heuristics; ship thin, iterate; timebox the needs engine
FERPA / pen test timing gates launch	High	Start compliance week 1; engage privacy counsel & pen-test vendor early
Roster connector / identity dedup complexity	Med	Confidence-scored matching + admin review queue; pilot with one clean source first
Accessibility treated as polish	High	Design a11y-in; axe gates in CI; audit before launch (special-education audience)
Key-person risk on the lead	Med	Document decisions (ADRs); pair on critical paths; cross-train developers